

MAT9004 - Assignment-3

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Q1, a) The loss function:

As the distance between the vehicle's location (x, y) and the known point (a_i, b_i) is given by:-

$$\sqrt{(x-a_i)^2 + (y-b_i)^2}$$

To find the approximated distances b/w the vehicle & the pt.

$$\text{loss} = \sqrt{(x-a_i)^2 + (y-b_i)^2} - d_i$$

Now the loss function $l(x, y)$, which measures the avg squared error over all the m known points is:-

$$l(x, y) = \frac{1}{m} \sum_{i=1}^m \left(\sqrt{(x-a_i)^2 + (y-b_i)^2} - d_i \right)^2$$

b) To find the gradient of the loss function $\nabla l(x, y)$

$$\nabla l(x, y) = \begin{bmatrix} l_x(x, y) \\ l_y(x, y) \end{bmatrix}$$

$\therefore$ the ~~the~~ partial derivatives with respect to x & y are:-

$$\begin{aligned} l_x(x, y) &= \frac{1}{m} \sum_{i=1}^m 2 \left(\sqrt{(x-a_i)^2 + (y-b_i)^2} - d_i \right) \times \frac{2(x-a_i)}{2\sqrt{(x-a_i)^2 + (y-b_i)^2}} \\ &= \frac{2}{m} \sum_{i=1}^m \left(\frac{x-a_i}{\sqrt{(x-a_i)^2 + (y-b_i)^2}} - d_i(x-a_i) \right) \end{aligned}$$

$$l_x(x, y) = \frac{2}{m} \sum_{i=1}^m \left((x - a_i) - \frac{\sum d_i (x - a_i)}{\sqrt{(x - a_i)^2 + (y - b_i)^2}} \right)$$

$\therefore$ Similarly:

$$l_y(x, y) = \frac{2}{m} \sum_{i=1}^m \left((y - b_i) - \frac{\sum d_i (y - b_i)}{\sqrt{(x - a_i)^2 + (y - b_i)^2}} \right)$$

$\therefore$ for gradient:

$$\nabla l(x, y) =$$

$$\left(\frac{2}{m} \sum_{i=1}^m \left((x - a_i) - \frac{d_i (x - a_i)}{\sqrt{(x - a_i)^2 + (y - b_i)^2}} \right), \frac{2}{m} \sum_{i=1}^m \left((y - b_i) - \frac{d_i (y - b_i)}{\sqrt{(x - a_i)^2 + (y - b_i)^2}} \right) \right)$$

Q2 Given $f(x,y) = -2x^2 + 2xy - 3y^2 - 5x + 5y + 7$

a) $\nabla f(x,y)$

To find the gradient we will find the partial derivative:

$$f_x(x,y) = -4x + 2y - 5$$

$$f_y(x,y) = 2x - 6y + 5$$

$$\nabla f(x,y) = (-4x + 2y - 5, 2x - 6y + 5)$$

b) To find the Hessian matrix $H(x,y)$ for f :

$$H(x,y) = \begin{bmatrix} f_{xx}(x,y) & f_{xy}(x,y) \\ f_{yx}(x,y) & f_{yy}(x,y) \end{bmatrix}$$

$$f_{xx}(x,y) = -4$$

$$f_{yy}(x,y) = -6$$

To complete the second partial derivative:-

$$f_{xx}(x,y) = -4$$

$$f_{yy}(x,y) = -6$$

$$f_{xy}(x,y) = +2$$

$$f_{yx}(x,y) = 2$$

$$\therefore H(x,y) = \begin{bmatrix} -4 & 2 \\ 2 & -6 \end{bmatrix}$$

c) Find and classify the stationary points:-

Gradient $\nabla f(x, y)$ should be zero:

$$\nabla f(x, y) = (-4x + 2y - 5, 2x - 6y + 5) = (0, 0)$$

This gives the system of ~~system~~ equations:

$$-4x + 2y - 5 = 0 \quad (1)$$

$$2x - 6y + 5 = 0 \quad (2)$$

$\Rightarrow$ From equation (1), solve for y :

$$2y = 4x + 5 \Rightarrow y = 2x + \frac{5}{2}$$

Using this into equation (2):

$$2x - 6\left(2x + \frac{5}{2}\right) + 5 = 0$$

Simplify: $2x - 12x - 15 + 5 = 0$

$$-10x - 10 = 0$$

$$x = -1$$

$\therefore$ y comes out to be: $y = -2 + \frac{5}{2}$

$$= \frac{1}{2}$$

$\therefore$ Stationary point is $\left(-1, \frac{1}{2}\right)$

Classification of the stationary point :-

$$H(x,y) = \begin{pmatrix} -4 & 2 \\ 2 & -6 \end{pmatrix}$$

The determinant of the Hessian is :-

$$\det(H) = (-4)(-6) - (2)(2) = 24 - 4 = 20$$

Since $\det(H) > 0$, the Hessian is +ve

meaning the stationary point $(-1, \frac{1}{2})$ is a local maximum.

e) Find the global maximum and global minimum of f on D
On the boundaries of D

• At $(0,0)$

$$f(0,0) = -2(0)^2 + 2(0)(0) - 3(0)^2 - 5(0) + 5(0) + 7 = 7$$

• At $(2,0)$

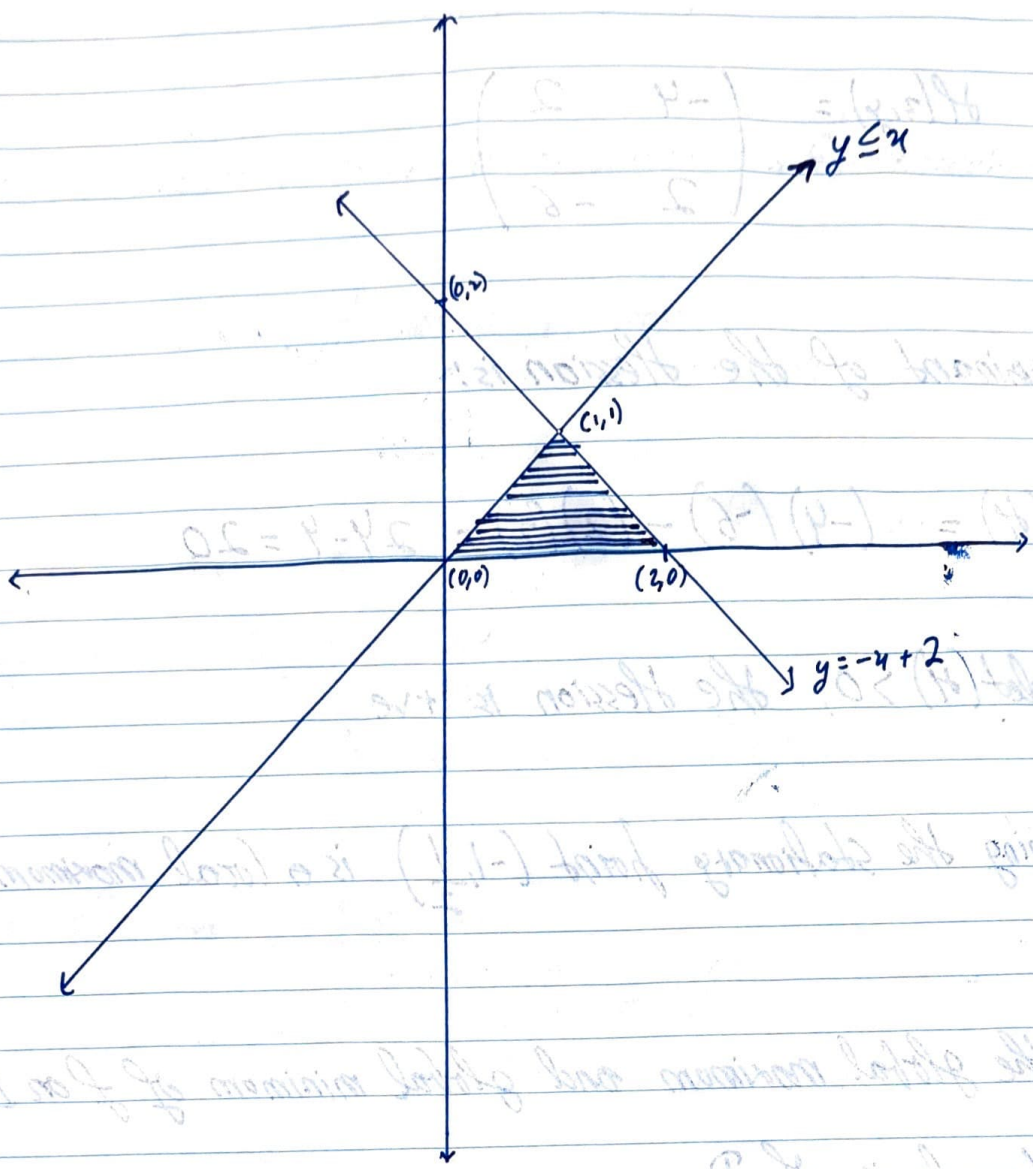
$$f(2,0) = -2(2)^2 + 2(2)(0) - 3(0)^2 - 5(2) + 5(0) + 7 = -8 - 10 + 7 = -11$$

• At $(1,1)$

$$f(1,1) = -2(1)^2 + 2(1)(1) - 3(1)^2 - 5(1) + 5(1) + 7 = -2 + 2 - 3 - 5 + 5 + 7 = 4$$

$\therefore$ global maximum is at $(0,0)$ & global minimum is at $(2,0)$

d.)



At $(0,0)$.

$$Z = 5 + (0)2 + (0)2 - 5(0) - 3(0) = 5$$

At $(2,0)$.

$$Z = 5 + (2)2 + (0)2 - 5(2) - 3(0) = 1$$

At $(1,1)$.

$$Z = 5 + (1)2 + (1)2 - 5(1) - 3(1) = 1$$

∴ Global minimum is at $(0,0)$ & Global maximum is at $(2,0)$.

3. a) No. of ways to create a single batch of 10 loaves

Kind of breads:- 5

batch size = 10

As repetition of the loaves are allowed this will be a combⁿ with repetition problem.

$$\therefore \text{formula} = \binom{n+r-1}{r-1} = \binom{n+r-1}{n}$$

Here $n = 5$

$r = 10$

$$\text{Number of ways} = \binom{10+5-1}{5-1} = 14$$

$$\binom{10+5-1}{5-1} = \binom{14}{4}$$

$$\Rightarrow \frac{14!}{4!10!} = \frac{14 \times 13 \times 12 \times 11}{4 \times 3 \times 2 \times 1} = 1001$$

b) Similarly:-

As kinds of savory bread = 3

kinds of sweet bread = 2

$$\Rightarrow \binom{10+3-1}{10} = \binom{12}{10} \Rightarrow 66$$

And for sweet buns

~~kinds of sweet bread~~

$$\binom{10+2-1}{10} = \binom{11}{10} = 11$$

$$\begin{aligned} \text{Total batch} &= 66 + 11 \\ &= 77 \end{aligned}$$

c) the problem will be solved with the permutation formula:
~~which will~~

As the total baking schedule is ^{for} 7 batches

Number of schedule will be for 7!

$\Rightarrow 7!$ is the way to arrange 7 batches.

d) Constraints:

- 1.) At most 4 loaves of the same kind of savoury bread in any savoury batch
- 2.) At least 4 loaves of each kind of sweet bread in any sweet batch
- 3.) The first two and last two batches must be different flavors.

$\therefore$ According to constraints:-

10 loaves should be distributed among 3 kind of savoury bread, with each kind limited to 4 loaves.

$\therefore$ the possible combination in that case will be:-

$$(4, 4, 2), (4, 3, 3)$$

$$\text{for } (4, 4, 2) \Rightarrow \frac{3!}{2!1!} = \frac{6}{2} = 3 \quad (\text{As 2 types are same we divide with } 2!)$$

$$\text{for } (4, 2, 3) \therefore \frac{3!}{1! 2!} = \frac{6}{2} = 3$$

$\therefore$ total ways of soroury batch is $\Rightarrow 6$

2) for sweet batch looks:

The combination according to constraints are:

$\Rightarrow (4, 6) (5, 5) (6, 4)$

$\therefore$ the total no. of ways $\Rightarrow 3$

Now putting the flavour constraints:-

The first two and last two batches must be different.

$\therefore$ the first and second batches could be:-

- $\Rightarrow$ Sweet, Soroury
- $\Rightarrow$ Soroury, Sweet

Similarly the last & second last batches could be:-

- $\Rightarrow$ Sweet, Soroury
- $\Rightarrow$ Soroury, Sweet

$\therefore$ there are 4 ways of arranging the flavours.

4a) To find the no. of possible schedules:-

As each customer is visited only once

it is a permutation problem :-

$$\text{No. of schedules} = 20!$$

b) Now for 2 drivers each doing 10 deliveries:-

the no. of ways to choose 10 customers is :-

$$\binom{20}{10}$$

And when checked for 2 drivers with 10 customers each

$$\binom{20}{10} \times 10! \times 10!$$

$\therefore$ This number is smaller than no. of schedules.

c) Adding additional constraints:-

$\Rightarrow$ 8 customers must be in the first 5 deliveries

$\Rightarrow$ 3 customers must be in the last 5 deliveries

$\therefore$ choose 5 deliveries for the first driver with 8 customers

$$\Rightarrow \binom{8}{5}$$

As the no. of ways to arrange 5 chosen customers is $5!$ and remaining can be chosen in $3!$

$$\Rightarrow \binom{8}{5} \times 5! \times 3!$$

Arranging
2) Removing customers for lost 4 drivers:-

$$\binom{4}{3}$$

2) Arranging the unrestricted customers.

As 9 customers are left, the need to be split among the two drivers:-

one gets 5 of them other gets 4

to assign a route $\Rightarrow \binom{9}{5}$

to arrange them in the slot is $5!$
and for other driver is $4!$

$\therefore$ total no. :-

$$\binom{8}{5} \times 5! \times 3! \times \binom{4}{3} \times \binom{9}{5} \times 5! \times 4! = (8 \times 5 \times 3 \times 4 \times 9 \times 5 \times 4) \times 9$$

S

Probability (non-defective) = 0.95 = (success)

Probability (defective) = 0.05 = (failure)

a) formula :- ~~$P = \frac{n!}{k!}$~~

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

now $n = 5$ (no. of light bulbs)

$k = 3$ (no. of non-defective bulbs)

$p = 0.95$ (prob of a bulb ~~not~~ being non-defective)

$1-p = 0.05$ (prob. of bulb being defective.)

$$P(X=3) = \binom{5}{3} (0.95)^3 (0.05)^2$$

$$P(X=3) = \frac{5!}{3!(5-3)!} (0.95)^3 (0.05)^2$$

$$P(K=3) = 10 (0.95)^3 (0.05)^2$$

$$= 0.0214$$

b) $\Rightarrow$ Prob. that all 6 bulbs are non-defective:-

$$P(\text{all 6 non-defective}) = (0.95)^6$$

$\Rightarrow$ Prob. that a pack contains at least one defective bulb:-

$$P(\text{at least one defective}) = 1 - (0.95)^6$$

$\therefore$ Expected no. of defective packs in a box of 20 packs:-

$$E(\text{defective packs}) = 20 \times P(\text{at least 1 defective})$$

$$E(\text{defective packs}) = 20 \times (1 - (0.95)^6)$$

c) Prob. of bulb being defective $P(\text{defective}) = 0.05$

Prob. of a bulb being non-defective = 0.95

For a pack to contain only non-defective bulbs, the prob. is:-

$$P(\text{all bulbs non-defective}) = 0.95^6$$

So, Prob. of at least one bulb is defective in a pack is

$$\begin{aligned} P(\text{at least 1 defective}) &= 1 - P(\text{all bulb non-defective}) \\ &= 1 - 0.95^6 \\ &= 1 - 0.7351 \approx 0.2649 \end{aligned}$$

Expected no. of defective packs (without testing): In a pack of 20 packs

$$20 \times 0.2649 \approx 5.298$$

Q6) According to the PDFs

$$P(X \geq 2) = \int_2^{\infty} f(z) dz$$

$$\therefore f(z) = \frac{3}{8} e^{1-z/2} \text{ for } z \geq 2$$

$\therefore$ Probability is

$$P(X \geq 2) = \int_2^{\infty} \frac{3}{8} e^{1-z/2} dz$$

$$= \frac{3}{8} \int_2^{\infty} e^{1-z/2} dz$$

$$= \frac{3}{8} \times 2e^{-1}$$

$$= \frac{3}{4e}$$

$$P(X \geq 2) \approx 12.850 - 1 =$$

$$SP2.2 \approx P(X \geq 2) \times 0.8$$

6) Probability that a single bulb operates for at least 2000 hrs

$$P(X \geq 2) \geq \frac{3}{4e}$$

→ now finding no. of bulbs to be installed :-

$n \propto P(X \geq 2)$ which needs to be at least 6

$$\therefore n \propto \frac{3}{4e} \geq 6$$

$$n \geq \frac{6 \times 4e}{3} = 8e \approx 21.8$$

$\therefore 22$ bulbs should be installed

now. Probability that five or fewer bulbs are still operating:-

we need binomial probability formulae

$$P(X \leq 5) = \sum_{k=0}^5 \binom{22}{k} p^k (1-p)^{22-k}$$

c) Expected lifetime of a bulb.

$$E[X] = \int_0^2 x f(x) dx + \int_2^{\infty} x f(x) dx \quad (E_1[X] + E_2[X])$$

$\therefore$ for $0 \leq x < 2$

$$f(x) = \frac{3}{32} x^2$$

$$E_1[X] = \int_0^2 x \cdot \frac{3}{32} x^2 dx = \frac{3}{32} \int_0^2 x^3 dx = \frac{3}{32} \times \frac{16}{4} = \underline{\underline{\frac{3}{8}}}$$

$\therefore$ for $x \geq 2$ $f(x) = \frac{3}{8} e^{1-\frac{x}{2}}$

$$E_2[X] = \int_2^{\infty} x \cdot \frac{3}{8} e^{1-\frac{x}{2}} dx$$

$$= \frac{3}{8} \times 6e^{-1} = \frac{9}{4e}$$

$$\therefore E[X] = E_1[X] + E_2[X]$$

$$= \frac{3}{8} + \frac{9}{4e}$$

d) $E(x)$ ~~den~~ where expected no. of failures = 1

$\therefore$

$$E(x) = n(1-h(T))$$

$[h(T) = \text{prob where bulb lasts longer than } T]$

$$1 = 6(1-h(T))$$

$$\therefore h(T) = \frac{5}{6}$$

$\therefore$ According to PDFs

$$h(T) = \int_2^+ f(z) dz$$

$$\frac{5}{6} = \int_2^+ \frac{3}{8} e^{1-\frac{z}{2}} dz$$

$$\frac{5}{6} = \frac{3}{8} e \int_2^+ e^{-\frac{z}{2}} dz$$

$$\frac{5}{6} = \frac{3}{8} e [e^{-\frac{z}{2}}]_2^+$$

$$= \frac{3}{8} e [e^{-\frac{+}{2}} - e^0]$$

$$-\frac{5}{6} \times \frac{8}{3e} = e^{-\frac{+}{2}} - 1$$

$$-\frac{20}{9e} = e^{-\frac{+}{2}} - 1 \quad \Rightarrow \quad e^{-\frac{+}{2}} = -\frac{20}{9e} + 1$$

$$+ = -2 \ln \left(-\frac{20}{9e} + 1 \right)$$

7) a) to find the flipping an unbiased coin variable we can check the marginal probabilities of X and Y

→ marginal probability of $X=0$ is

$$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) = 0.2 + 0.1 = 0.3$$

$$P(X=1) = P(X=1, Y=0) + P(X=1, Y=1) = 0.3 + 0.4 = 0.7$$

Similarly ~~then~~ $P(Y=0) = 0.5$ & $P(Y=1) = 0.5$

∴ ~~Y~~ corresponds to flipping an unbiased coin.

b) to check if X & Y are independent they must satisfy

$$P(X=x, Y=y) = P(X=x)P(Y=y) \text{ for all } x \text{ & } y$$

$$P(X=0) = 0.3$$

$$P(X=1) = 0.7$$

$$P(Y=0) = 0.5$$

$$P(Y=1) = 0.5$$

$$\therefore P(X=0, Y=0) = 0.2$$

$$P(X=0) = 0.3 \text{ & } P(Y=0) = 0.5$$

$$0.2 \neq 0.15$$

∴ they are not independent.

$$a) E[XY] = \sum_{x,y} x \cdot y \cdot P(X=x, Y=y)$$

$$= 0.2(0)(0) + 0.3(1)(0) + (0.1)(0)(1) + (0.4)(1)(1)$$

$$= \underline{0.4}$$

$$E[Var[XY]] = E[(XY)^2] - (E[XY])^2$$

$$= 0.4 - (0.4)^2$$

$$= 0.4 - 0.16$$

$$= \underline{0.24}$$